

Probability Recitation

Sophia Stiles

CS155 Machine Learning and Data Mining

February 20, 2026

Motivation

- Uncertainty is everywhere around us

Motivation

- Uncertainty is everywhere around us
 - “what is the chance that it will rain today?”

Motivation

- Uncertainty is everywhere around us
 - “what is the chance that it will rain today?”
 - “when will the next bus arrive?”

Motivation

- Uncertainty is everywhere around us
 - “what is the chance that it will rain today?”
 - “when will the next bus arrive?”
 - “will I go to recitation today?”

Motivation

- Uncertainty is everywhere around us
 - “what is the chance that it will rain today?”
 - “when will the next bus arrive?”
 - “will I go to recitation today?”
- Machine learning tries to understand uncertainties and interact with the real world

Motivation

- Uncertainty is everywhere around us
 - “what is the chance that it will rain today?”
 - “when will the next bus arrive?”
 - “will I go to recitation today?”
- Machine learning tries to understand uncertainties and interact with the real world
- Probability theory is the mathematical study of uncertainty

Basic Concepts

- Sample Space Ω : set of all possible outcomes

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$
 - $\mathbb{P}(\Omega) = 1$

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$
 - $\mathbb{P}(\Omega) = 1$
 - $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$
 - $\mathbb{P}(\Omega) = 1$
 - $\mathbb{P}(\bar{A}) = 1 - \mathbb{P}(A)$

Examples

Suppose you are rolling a six-sided die. Then we have:

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$
 - $\mathbb{P}(\Omega) = 1$
 - $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

Examples

Suppose you are rolling a six-sided die. Then we have:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$
 - $\mathbb{P}(\Omega) = 1$
 - $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

Examples

Suppose you are rolling a six-sided die. Then we have:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{6\}) = \frac{1}{6}$

Basic Concepts

- Sample Space Ω : set of all possible outcomes
- An event, A , is a subset of Ω (events form a σ -algebra)
- $\mathbb{P}(A)$ is the probability of event A occurring (measure over $(\Omega, \mathcal{A})$)
 - $0 \leq \mathbb{P}(A) \leq 1$
 - $\mathbb{P}(\emptyset) = 0$
 - $\mathbb{P}(\Omega) = 1$
 - $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

Examples

Suppose you are rolling a six-sided die. Then we have:

- $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{6\}) = \frac{1}{6}$
- $\mathbb{P}(\{1, 2, 3, 4, 5, 6\}) = 1$

Given two events, A and B , the probability of A or B is

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \quad (1)$$

$$= \mathbb{P}(A) + \mathbb{P}(B) \quad (2)$$

where (1) and (2) are equal only if A and B are mutually exclusive

Examples

Suppose you are rolling a six-sided die. Then we have:

Given two events, A and B , the probability of A or B is

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \quad (1)$$

$$= \mathbb{P}(A) + \mathbb{P}(B) \quad (2)$$

where (1) and (2) are equal only if A and B are mutually exclusive

Examples

Suppose you are rolling a six-sided die. Then we have:

- $\mathbb{P}(\{2, 4, 6\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{1}{2}$

Given two events, A and B , the probability of A or B is

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \quad (1)$$

$$= \mathbb{P}(A) + \mathbb{P}(B) \quad (2)$$

where (1) and (2) are equal only if A and B are mutually exclusive

Examples

Suppose you are rolling a six-sided die. Then we have:

- $\mathbb{P}(\{2, 4, 6\}) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{1}{2}$
- $\mathbb{P}(\{1, 2, 3\} \cup \{2, 4, 6\}) =$
 $\mathbb{P}(\{1, 2, 3\}) + \mathbb{P}(\{2, 4, 6\}) - \mathbb{P}(\{2\}) = \frac{1}{2} + \frac{1}{2} - \frac{1}{6} = \frac{5}{6}$

Joint and Conditional Probabilities

Given two events, A and B ,

- $\mathbb{P}(A, B)$ is the joint probability of A and B occurring together
- $\mathbb{P}(A \mid B)$ is the conditional probability of A occurring given that we know B has occurred

Joint and Conditional Probabilities are related in the following way:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A, B)}{\mathbb{P}(B)}$$

where we assume that $\mathbb{P}(B) \neq 0$.

Joint and Conditional Probabilities

Examples

Consider a standard deck of cards. What is the probability that the first two cards drawn are both Kings?

- Event A = First card drawn is a King
- Event B = Second card drawn is a King

A standard deck of cards has 4 Kings and 52 total cards.

Joint and Conditional Probabilities

Examples

Consider a standard deck of cards. What is the probability that the first two cards drawn are both Kings?

- Event A = First card drawn is a King
- Event B = Second card drawn is a King

A standard deck of cards has 4 Kings and 52 total cards.

- $\mathbb{P}(A) = \frac{4}{52}$

Joint and Conditional Probabilities

Examples

Consider a standard deck of cards. What is the probability that the first two cards drawn are both Kings?

- Event A = First card drawn is a King
- Event B = Second card drawn is a King

A standard deck of cards has 4 Kings and 52 total cards.

- $\mathbb{P}(A) = \frac{4}{52}$
- $\mathbb{P}(B \mid A) = \frac{3}{51}$

Joint and Conditional Probabilities

Examples

Consider a standard deck of cards. What is the probability that the first two cards drawn are both Kings?

- Event A = First card drawn is a King
- Event B = Second card drawn is a King

A standard deck of cards has 4 Kings and 52 total cards.

- $\mathbb{P}(A) = \frac{4}{52}$
- $\mathbb{P}(B | A) = \frac{3}{51}$

This means that

- $\mathbb{P}(A, B) = \mathbb{P}(B | A)\mathbb{P}(A) = \frac{3}{51} \cdot \frac{4}{52} = \frac{1}{221} \approx 0.005$

Joint Probabilities - Chain Rule

An extension of $\mathbb{P}(A, B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$:

$$\begin{aligned}\mathbb{P}(A_1, A_2, \dots, A_n) &= \mathbb{P}(A_n, \dots, A_2, A_1) \\ &= \mathbb{P}(A_n \mid A_{n-1}, \dots, A_2, A_1) \mathbb{P}(A_{n-1}, \dots, A_2, A_1) \\ &\quad \vdots \\ &= \prod_{i=1}^n \mathbb{P}(A_i \mid A_1, A_2, \dots, A_{i-1})\end{aligned}$$

Independence

Two events, A and B , are independent if

$$\mathbb{P}(A, B) = \mathbb{P}(A)\mathbb{P}(B)$$

or equivalently

$$\mathbb{P}(A \mid B) = \mathbb{P}(A)$$

In other words, knowledge of whether B occurred does not affect the probability that A occurs

Examples

Suppose that you roll a six-sided die twice. What is the probability that you roll 1 both times?

- A = rolling a 1 in the first roll
- B = rolling a 1 in the second roll

These two events are independent so

$$\mathbb{P}(A, B) = \mathbb{P}(A)\mathbb{P}(B) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Bayes Theorem

We know that the following is true

$$\mathbb{P}(A, B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$$

This implies that

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

$$\mathbb{P}(A \mid B) \propto \mathbb{P}(B \mid A)\mathbb{P}(A)$$

Definitions

Suppose we wish to examine some event A (usually data) that depends on some event B (usually parameters)

Suppose we wish to examine some event A (usually data) that depends on some event B (usually parameters)

- $\mathbb{P}(A)$ - Prior Information

The probability of A *before* we know anything about B .

Definitions

Suppose we wish to examine some event A (usually data) that depends on some event B (usually parameters)

- $\mathbb{P}(A)$ - Prior Information
The probability of A *before* we know anything about B .
- $\mathbb{P}(A \mid B)$ - Posterior Probability
The probability of A *after* we have observed B .

Definitions

Suppose we wish to examine some event A (usually data) that depends on some event B (usually parameters)

- $\mathbb{P}(A)$ - Prior Information
The probability of A *before* we know anything about B .
- $\mathbb{P}(A \mid B)$ - Posterior Probability
The probability of A *after* we have observed B .
- $\mathbb{L}(B \mid A) = \mathbb{P}(A \mid B)$ - Likelihood Function
The likelihood of B after observing A .

Suppose we wish to examine some event A (usually data) that depends on some event B (usually parameters)

- $\mathbb{P}(A)$ - Prior Information
The probability of A *before* we know anything about B .
- $\mathbb{P}(A \mid B)$ - Posterior Probability
The probability of A *after* we have observed B .
- $\mathbb{L}(B \mid A) = \mathbb{P}(A \mid B)$ - Likelihood Function
The likelihood of B after observing A .
- $\mathbb{P}(B)$ - Evidence
The probability that our condition is true

Examples

If a person has an allergy (A), sneezing (S) is observed with probability $\mathbb{P}(S | A) = 0.8$. What is the chance a person has an allergy given that they are sneezing: $\mathbb{P}(A | S)$?

- Assume that $\mathbb{P}(A) = 0.001$ (few people have allergies)
- Assume that $\mathbb{P}(S) = 0.1$ (many people sneeze).

$$\begin{aligned}\mathbb{P}(A | S) &= \frac{\mathbb{P}(S | A)\mathbb{P}(A)}{\mathbb{P}(S)} \\ &= \frac{0.8 \cdot 0.001}{0.1} \\ &= 0.008\end{aligned}$$

Random Variables

So far, we have only considered simple examples with simple events as the possible outcomes

- $A =$ rolling 1 on a six-sided dice
- $B =$ first card drawn is a King

Random Variables

So far, we have only considered simple examples with simple events as the possible outcomes

- A = rolling 1 on a six-sided dice
- B = first card drawn is a King

This is mathematically imprecise and can be limiting. The notion of random variables resolves these issues

- A random variable X is a function $X : \Omega \rightarrow \mathbb{R}$
- Abuse of notation: $\mathbb{P}(X = x)$ is the probability that the random variable takes on value x

Random Variables

Examples

- Dice Roll: $X = i$ for $i \in \{1, 2, 3, 4, 5, 6\}$ with $\mathbb{P}(X = i) = \frac{1}{6}$

Random Variables

Examples

- Dice Roll: $X = i$ for $i \in \{1, 2, 3, 4, 5, 6\}$ with $\mathbb{P}(X = i) = \frac{1}{6}$
- Biased Coin: $X = 1$ with $\mathbb{P}(X = 1) = p$ and $X = 0$ with $\mathbb{P}(X = 0) = 1 - p$

Random Variables

Examples

- Dice Roll: $X = i$ for $i \in \{1, 2, 3, 4, 5, 6\}$ with $\mathbb{P}(X = i) = \frac{1}{6}$
- Biased Coin: $X = 1$ with $\mathbb{P}(X = 1) = p$ and $X = 0$ with $\mathbb{P}(X = 0) = 1 - p$

All random variables have a Cumulative Distribution Function (CDF):

$$F(x) = \mathbb{P}(X \leq x)$$

Some properties of the CDF are:

- $0 \leq F(x) \leq 1$
- $F(X)$ is monotonically increasing

$$\lim_{x \rightarrow -\infty} F(x) = 0$$

$$\lim_{x \rightarrow \infty} F(x) = 1$$

Discrete Random Variables

A random variable that can take on only a finite (or countably infinite) number of different values

Discrete random variables have a Probability Mass Function (PMF):

$$p(x) = \mathbb{P}(X = x)$$

The PMF satisfies

$$\sum_x \mathbb{P}(X = x) = 1$$

Discrete Random Variables

A random variable that can take on only a finite (or countably infinite) number of different values

Discrete random variables have a Probability Mass Function (PMF):

$$p(x) = \mathbb{P}(X = x)$$

The PMF satisfies

$$\sum_x \mathbb{P}(X = x) = 1$$

Examples

Suppose you are flipping two coins. Let X be the random variable that equals the number of heads

$\mathbb{P}(X = 0)$	$\mathbb{P}(X = 1)$	$\mathbb{P}(X = 2)$
0.25	0.5	0.25

Continuous Random Variables

A random variable for which almost never takes on (measure zero) any particular value; formally, continuous CDF.

Continuous random variables have a Probability Distribution Function (PDF):

$$p(x) = f(x) = \frac{d}{dx}F(x)$$

The PDF satisfies

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Continuous Random Variables

A random variable for which almost never takes on (measure zero) any particular value; formally, continuous CDF.

Continuous random variables have a Probability Distribution Function (PDF):

$$p(x) = f(x) = \frac{d}{dx}F(x)$$

The PDF satisfies

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

Examples

Suppose that X is a random variable that is equal to a value chosen uniformly at random from the interval $[0, a]$. Then

$$F(x) = \frac{x}{a} \qquad f(x) = \frac{1}{a}$$

Marginal Distribution

Suppose that for two random variables, X and Y , the joint distribution $p(x, y)$ is known for all combinations of X and Y . Then the marginal distribution of X is

$$p(x) = \sum_y p(x, y) \qquad p(x) = \int_{-\infty}^{\infty} p(x, y) dy$$

and the marginal distribution of Y is

$$p(y) = \sum_x p(x, y) \qquad p(y) = \int_{-\infty}^{\infty} p(x, y) dx$$

Marginal Distribution

Examples

Consider two random variables X and Y . Suppose the joint distribution is given by

	x_1	x_2	x_3	x_4	$p(y)$
y_1	$\frac{4}{32}$	$\frac{2}{32}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{8}{32}$
y_2	$\frac{2}{32}$	$\frac{4}{32}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{8}{32}$
y_3	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{8}{32}$
y_4	$\frac{8}{32}$	0	0	0	$\frac{8}{32}$
$p(x)$	$\frac{16}{32}$	$\frac{8}{32}$	$\frac{4}{32}$	$\frac{4}{32}$	$\frac{32}{32}$

$$p(x_1) = \sum_y p(x_1, y) = \frac{16}{32}$$

Expected Value

The expected value of a random variable, X , is the distribution-weighted mean

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)$$

$$\mathbb{E}[X] = \int_x xf(x)dx$$

Expected Value

The expected value of a random variable, X , is the distribution-weighted mean

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)$$

$$\mathbb{E}[X] = \int_x xf(x)dx$$

Expectation is linear

- $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$
- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Expected Value

The expected value of a random variable, X , is the distribution-weighted mean

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x)$$

$$\mathbb{E}[X] = \int_x xf(x)dx$$

Expectation is linear

- $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$
- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Examples

Suppose that X and Y are random variables that are 1 with probability p and 0 otherwise. What is the expected value of $X + Y$?

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] = \mathbb{P}(X = 1) + \mathbb{P}(Y = 1) = 2p$$

Variance

The variance of a random variable is the measure of the 'spread' of the values the variable takes on:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

The variance can also be written as

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Variance

The variance of a random variable is the measure of the 'spread' of the values the variable takes on:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

The variance can also be written as

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Variance is not linear

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

Covariance

The covariance between random variables X and Y measures the degree to which X and Y are related

$$\text{Cov}(X, Y) = \mathbb{E} \left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) \right]$$

This can also be written as

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Covariance

The covariance between random variables X and Y measures the degree to which X and Y are related

$$\text{Cov}(X, Y) = \mathbb{E} \left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) \right]$$

This can also be written as

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Note that $\text{Cov}(X, X) = \text{Var}(X)$

Variance and Covariance

Examples

Suppose that X and Y are random variables that are 1 with probability p and 0 otherwise. What is the covariance of X and Y ? What is the variance of $X + Y$?

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \\ &= p^2(1) + (1 - p^2)(0) - p^2 = 0\end{aligned}$$

$$\begin{aligned}\text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 + \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \\ &= p - p^2 + p - p^2 = 2p(1 - p)\end{aligned}$$

Important Discrete Distributions

- Bernoulli (p) - flipping a biased coin that lands heads with probability p , x is 1 for heads and 0 for tails
 - $\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$ or $\mathbb{P}(0) = 1 - p$ and $\mathbb{P}(1) = p$
 - $\mathbb{E}[x] = p$

Important Discrete Distributions

- Bernoulli (p) - flipping a biased coin that lands heads with probability p , x is 1 for heads and 0 for tails
 - $\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$ or $\mathbb{P}(0) = 1 - p$ and $\mathbb{P}(1) = p$
 - $\mathbb{E}[x] = p$
- Binomial (n, p) - flipping n biased coins that each land heads with probability p , x is the number of heads
 - $\mathbb{P}(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$
 - $\mathbb{E}[x] = np$

Important Discrete Distributions

- Bernoulli (p) - flipping a biased coin that lands heads with probability p , x is 1 for heads and 0 for tails
 - $\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$ or $\mathbb{P}(0) = 1 - p$ and $\mathbb{P}(1) = p$
 - $\mathbb{E}[x] = p$
- Binomial (n, p) - flipping n biased coins that each land heads with probability p , x is the number of heads
 - $\mathbb{P}(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$
 - $\mathbb{E}[x] = np$
- Geometric (p) - flipping a biased coin that lands heads with probability p , until it lands on heads, x is the number of tosses
 - $\mathbb{P}(X = x) = p(1 - p)^{x-1}$
 - $\mathbb{E}[x] = \frac{1}{p}$

Important Discrete Distributions

- Bernoulli (p) - flipping a biased coin that lands heads with probability p , x is 1 for heads and 0 for tails
 - $\mathbb{P}(X = x) = p^x(1 - p)^{1-x}$ or $\mathbb{P}(0) = 1 - p$ and $\mathbb{P}(1) = p$
 - $\mathbb{E}[x] = p$
- Binomial (n, p) - flipping n biased coins that each land heads with probability p , x is the number of heads
 - $\mathbb{P}(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$
 - $\mathbb{E}[x] = np$
- Geometric (p) - flipping a biased coin that lands heads with probability p , until it lands on heads, x is the number of tosses
 - $\mathbb{P}(X = x) = p(1 - p)^{x-1}$
 - $\mathbb{E}[x] = \frac{1}{p}$
- Poisson (λ) - receiving letters when the average number of letters received per day is λ , x is the number of letters received in a given day
 - $\mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$
 - $\mathbb{E}[x] = \lambda$

Important Continuous Distributions

- Uniform (a, b) with $a \leq b$ - spinning a board game spinner with uniform values from a to b , x is the resultant value
 - $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$, 0 otherwise
 - $\mathbb{E}[x] = \frac{a+b}{2}$

Important Continuous Distributions

- Uniform (a, b) with $a \leq b$ - spinning a board game spinner with uniform values from a to b , x is the resultant value
 - $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$, 0 otherwise
 - $\mathbb{E}[x] = \frac{a+b}{2}$
- Normal (μ, σ^2) - heights in a general population with average μ and variance σ^2 , x is the height of some person in that population
 - $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$
 - $\mathbb{E}(x) = \mu$

Law of Large Numbers

Let $X_1, X_2, \dots, X_n$ be a set of independent and identically distributed (iid) random variables with $\mathbb{E}[X_i] = \mu$. Then the sample average given by

$$\bar{X}_n = \frac{1}{n}(X_1 + X_1 + \dots + X_n)$$

converges to the expected value

$$\bar{X}_n \rightarrow \mu \text{ as } n \rightarrow \infty$$

Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be a set of independent and identically distributed (iid) random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$, both finite. If we define the random variable Y as

$$Y = \frac{1}{n}(X_1 + X_1 + \dots + X_n)$$

then Y is approximately normal with mean μ and variance $\frac{\sigma^2}{n}$

The approximation improves as $n \rightarrow \infty$

Independent and Identically Distributed

The importance of the iid assumption cannot be overstated!

Independence assumption simplifies many things:

$$\mathbb{P}(X, Y) = \mathbb{P}(X)\mathbb{P}(Y)$$

$$\mathbb{P}(X \mid Y) = \mathbb{P}(X)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

We'd like to detect (discrete) or estimate (generally continuous).

- Bayesian vs frequentist perspective of inference.
- We'll describe a common form of frequentist inference relevant for CS155: maximum-likelihood-estimation.

Parameter Estimation

Suppose we have a parametrized distribution $\mathbb{P}(X|\theta)$ with θ unknown

Parameter Estimation

Suppose we have a parametrized distribution $\mathbb{P}(X|\theta)$ with θ unknown

Suppose we have an iid set of samples $x_1, \dots, x_n$

Parameter Estimation

Suppose we have a parametrized distribution $\mathbb{P}(X|\theta)$ with θ unknown

Suppose we have an iid set of samples $x_1, \dots, x_n$

How can we estimate θ ? From Bayes' Theorem, we have that

$$\mathbb{P}(\theta | X) \propto \mathbb{P}(X | \theta)\mathbb{P}(\theta)$$

- MAP: $\hat{\theta} = \arg \max_{\theta} \mathbb{P}(\theta | X)$
- MLE: $\hat{\theta} = \arg \max_{\theta} \mathbb{L}(\theta | X) = \arg \max_{\theta} \mathbb{P}(X | \theta)$

Parameter Estimate - log trick

The logarithmic function is monotonically increasing - will not change the location of where a function achieves its maximum

Multiplication turns into summation, resulting in less overflow

Considering the negative of the log likelihood function allows us to use gradient descent to minimize the function

$$\arg \max_{\theta} f(X | \theta) = \arg \min_{\theta} -\log f(X | \theta)$$

Parameter Estimation - Binomial Distribution

Examples

Suppose you toss a (possibly biased) coin n times and record the number of heads, k . What is p , the probability that it lands heads on a given coin flip. We can use the binomial distribution and MLE to find p :

$$\begin{aligned}\hat{p} &= \arg \max_p \mathbb{P}(k \mid p) = \arg \max_p \binom{n}{k} p^k (1-p)^{n-k} \\ &= \arg \max_p p^k (1-p)^{n-k} \\ &= \arg \min_p -\log p^k (1-p)^{n-k} \\ &= \arg \min_p -k \log p - (n-k) \log (1-p)\end{aligned}$$

Taking the derivative wrt to p and zeroing implies that $\hat{p} = \frac{k}{n}$

MLE with Logistic Regression

Examples

Suppose you have an iid dataset, $S = \{(x_i, y_i)_{i=1}^N\}$, and want to model it using the following model:

$$\mathbb{P}(y \mid x, w, b) = \frac{1}{1 + e^{-y(w^T x + b)}}$$

We can use a MLE to find w, b :

$$\hat{w}, \hat{b} = \arg \max_{w, b} \mathbb{P}(y \mid x, w, b)$$

$$= \arg \max_{w, b} \prod_{i=1}^N \mathbb{P}(y_i \mid x_i, w, b) \quad \text{iid assumption}$$

$$= \arg \min_{w, b} \sum_{i=1}^N -\ln \mathbb{P}(y_i \mid x_i, w, b) \quad \text{log-loss in logistic reg.}$$

References

- Lucy Yin's Recitation on Probability (CS155 Winter 2016)
- Kevin Murphy's Machine Learning: A Probabilistic Perspective
- Avishek Dutta's 2017 recitation on this topic